The chemistry of transition metals and Co-ordination Compounds.

Saif-50

[Farria-50]

TOPIC NAME :

DAY :

TIME :

DATE : / /

Basics

- **Complex compound:**

A complex compound can be defined as a class of substances with a chemical structure where the central atom is a metal and it is surrounded by non-metal atoms or a group of atoms or ligands.

Complex compound

Double salt

[এই complex এর প্রতিটি ion aq. soln এ ionized অবস্থায় থাকে। যেমনঃ $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O \xrightarrow{H_2O} Fe^{2+}, SO_4^{2-}, NH_4^+$]

True Complex

[এটির সবগুলো part-ই aq soln এ ionized হয় না। কোনো না কোনো অংশ unionized অবস্থায় থাকায় normal ion test এ শনাক্ত করা যায় না]

Normal complex

[এদের solid ও aq. structure দুটি eq. এ থাকে। Ex: $K_2[Cd(CN)_4] \rightleftharpoons 2K^+ + [Cd(CN)_4]^{2-} \rightleftharpoons 2K^+ + Cd^{2+} + 4CN^-$]

- $CoCl_3 \cdot 6NH_3$, $CoCl_3 \cdot 5NH_3 \cdot H_2O$ যৌগগুলোতে NH_3 থাকলেও এগুলোর basic character থাকে না। NH_3 মূলত নিজের lone pair e^- এর কারনে base হিসেবে কাজ করে। কিন্তু এক্ষেত্রে lone pair দিয়ে Co^{3+} আয়নের সাথে co-ordination bond গঠনের কারনে কোনো lone pair থাকে না। ও কারনে এটি কোনো acid এর সাথেও neutralization reaction দেয় না।

TOPIC NAME : _____ DAY : _____ TIME : DATE : /

• Warner's Theory :-

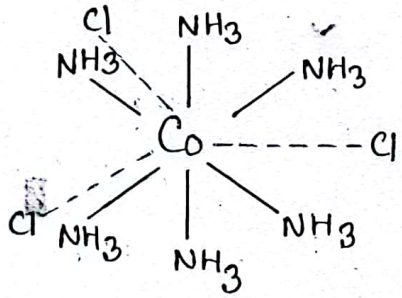
Valancy

- Primary
  - *only it ionizes
  - Satisfy only neg ions
- Secondary
  - Satisfy both neg ion and neutrals

** Co-ordination number is satisfied by secondary valancy.

- $CoCl_3 . 6NH_3$

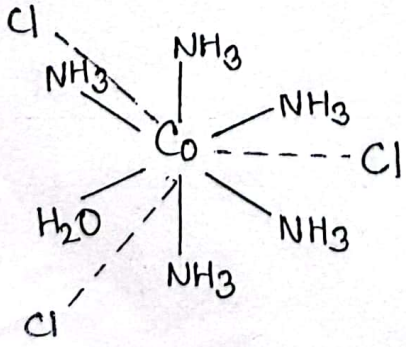
Here the Co-ordination number of Co^{3+} is = 6, which is satisfied by six NH_3. Then three Cl^- ions are connected via primary valancy.



• Only three Cl^- ionizes as anion.

- $CoCl_3 . 5NH_3 . H_2O$

Here Co has a coordination number 6 which is satisfied by 5 NH_3 and one H_2O. Three Cl^- ions satisfy P.V.



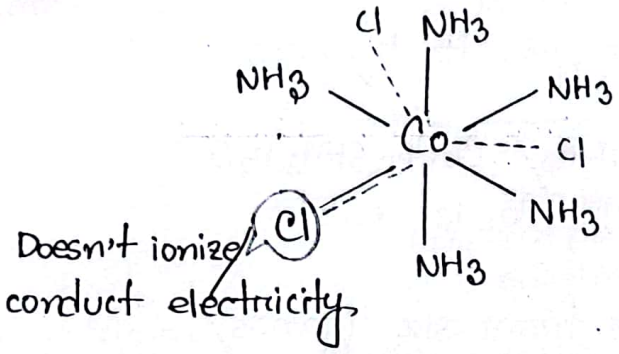
• Three Cl^- ionizes

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

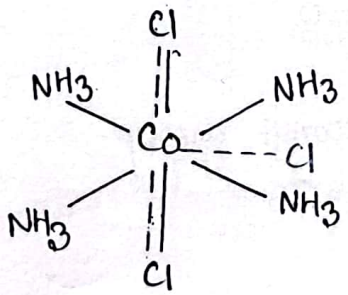
- $CoCl_3 . 5NH_3$:

Here 5 NH_3 satisfies 5 secondary valancy and one is satisfied by chlorine. Thus one chlorine satisfy both primary and secondary valanced. Mixed valanced ions can not be ionized. Thus only two Cl^- ions ionise.



- $CoCl_3.4NH_3$:

Six valancy should be fulfilled : of which four from NH_3 and rest two from two chlorine.



- Only one chlorine ionizes.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

• Effective Atomic Number: (EAN)

The number of electrons surrounding the nucleus of a metal atom in a metal complex is called EAN.

** If the EAN of any metal complex is the atomic number of any inert gas, it becomes stable.

✓ $CoCl_3.6NH_3$ / $CoCl_3.4NH_3$ / $CoCl_3.5NH_3.H_2O$

Here, oxidation state of Co is Co^{3+}.

Co^{3+} has 24 e^-.

Six pairs of electrons from six ligands / secondary valancy.

Total electrons = $24 + (6 \times 2) = 36$; [Kyptons atomic no.]

Thus it's a stable complex.

✓ $Fe(CO)_5$

Oxidation no of Fe = 0

Electron of Fe = 26

Two e^- from each carbonyl group.

Total e^- = $26 + (5 \times 2) = 36$

✓ $Cr(CO)_6$

oxidation no of Cr = 0

Total e^- = $24 + 12 = 36$

✓ $K_4[Fe(CN)_6]$

Oxidation no of Fe = +2

Electron of Fe^{2+} = $26 - 2 = 24$

Two e from each CN^-.

Total electron = $24 + 6 \times 2 = 36$

✓ $Cr(NH_3)_5Cl_3$

Oxidation no of Cr = +3

Two from each NH_3 and two from Cl.

Total e^- = $(24-3) + (5 \times 2) + (1 \times 2) = 38$.

unstable

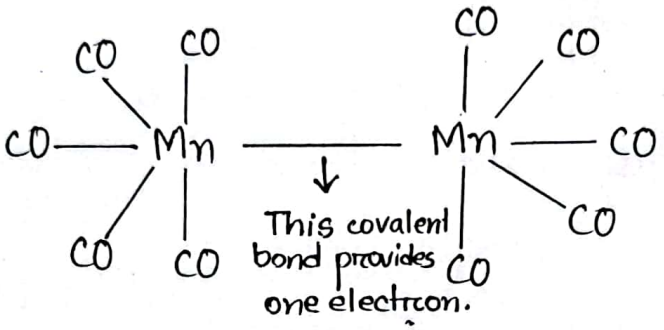
TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

• $Mn_2(CO)_5$ form dimer to satisfy stable configuration.

Oxidation no of Mn = 0

Total e^- = 25 + 2X5 = 35 , As it has one less electron tha. the neighbouring inert gas Kr, it forms dimer in order to get one electron from another Mn and to be stable



* Every carbonyl compound (C=O) follows EAN.

(*) $Fe_2(CO)_9$

$(CO)_4Fe - Fe(CO)_4$

C=O → Nature of this CO is different.

• Ligands/ Naming/ chelates from sheet

TOPIC NAME : _____ DAY : _____ TIME : DATE : /

• Formation of Co-ordination complex via VBT :-

(i) Salt dissociates to form metal ion M+.

(ii) Hybridization occures according to the electron of metal ion and incoming ion/ligand.

(iii) Re-arrangement occurs [if necessary].

(iv) Overlap with ligand.

(v) Geometry would be fixed.

* For $ns^2np^3nd^2 \longrightarrow sp^3d^2$

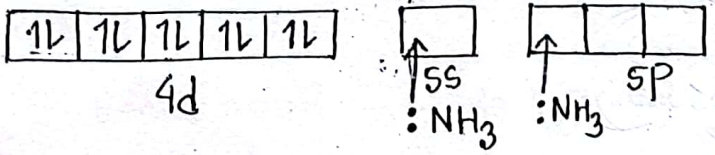
* For $ns^2np^3(n-1)d^2 \longrightarrow d^2sp^3$

• Some co-ordination complexes & their geometry:

• $[Ag(NH_3)_2]^+ \rightarrow Ag^+$

Ag (47) : [Kr] $4d^{10}5s^1$

Ag^+ :



As, 2 NH_3 would form complex, one pair of electron occupies 5S and another pair occupies 5P orbital and SP hybridization takes place.

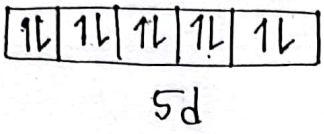
∴ Linear molecule & diamagnetic with no unpaired electron.

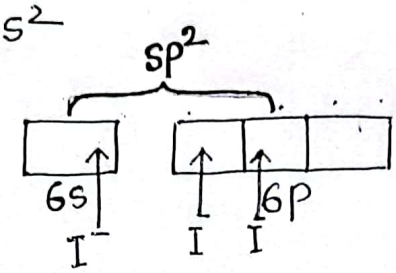
TOPIC NAME : _____ DAY : _____ TIME : DATE : / /

• $[HgI_3]^- \rightarrow Hg^{2+}$

$Hg\ (80) : [Xe]\ 4f^{14}\ 5d^{10}\ 6s^2$

Hg^{2+} :



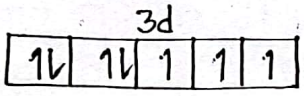


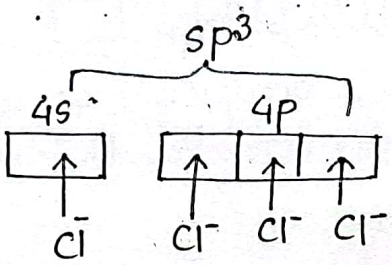
∴ Trigonal planar & diamagnetic.

• $[CoCl_4]^{2-} \rightarrow Co^{2+}$

$Co\ (27) : [Ar]\ 3d^7\ 4s^2$

Co^{2+} :

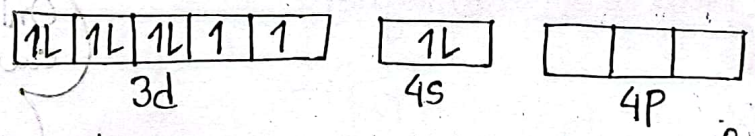




∴ Tetrahedral & paramagnetic.

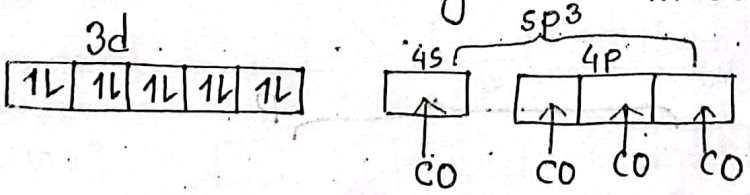
* ⊙ $[Ni(CO)_4] \rightarrow Ni$

$Ni\ (28) : [Ar]\ 3d^8\ 4s^2$



But there is not enough space to occupy four CO group. So, some sort of re-arrangement must be done.

Ni



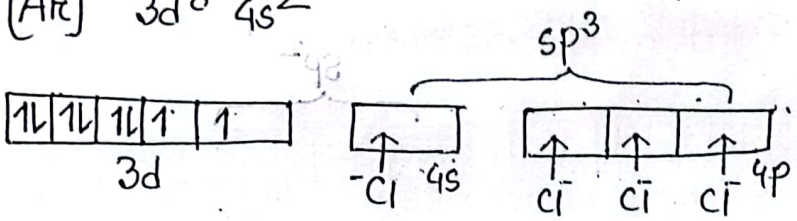
∴ Tetrahedral & diamagnetic.

TOPIC NAME : _____ DAY : _____ TIME : DATE : /

⊙ $[NiCl_4]^{2-} \rightarrow Ni^{2+}$

Ni (28) : [Ar] $3d^8\ 4s^2$

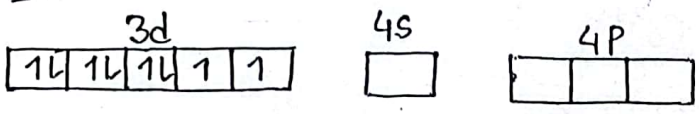
Ni^{2+} :



∴ Tetrahedral & paramagnetic

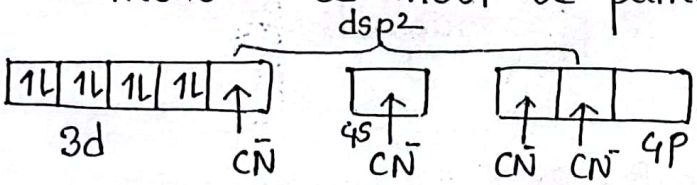
⊙ $[Ni(CN)_4]^{2-} \rightarrow Ni^{2+}$

Ni^{2+} :



If, four CN^- occupies 4s and three 4p orbital it remains paramagnetic. But according to experimental analysis, this complex acts a diamagnetic one. Thus the unpaired electrons of 3d must be paired.

Ni^{2+} :



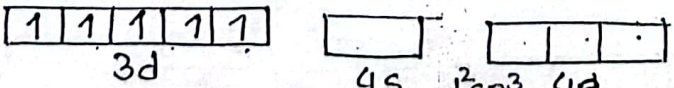
∴ Square planar & diamagnetic :

Only strong ligand can cause re-arrangement. Not the weak ones

⊙ $[Fe(CN)_6]^{3-} \rightarrow Fe^{3+} / [FeF_6]^{3-}$

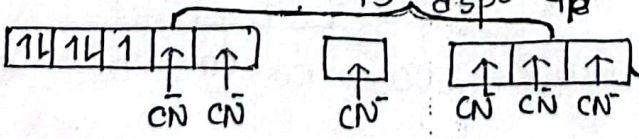
Fe (26) : [Ar] $3d^6\ 4s^2$

Fe^{3+} :



[Not enough space for six CN^-]

Fe^{3+} :



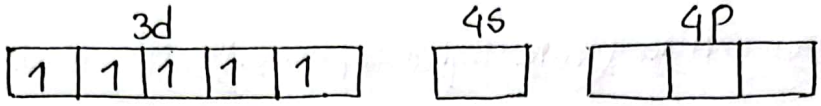
[CN^- is strong Ligand] → [Re-arrangement]

∴ Octahedral and paramagnetic :

TOPIC NAME : _____ DAY : _____ TIME : DATE : / /

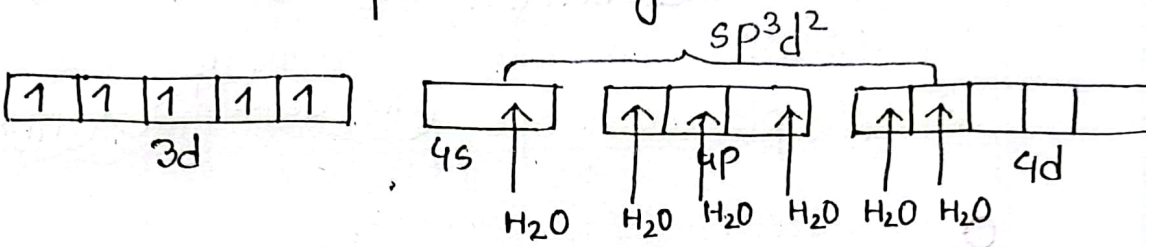
- $[Fe(H_2O)_6]^{3+} \rightarrow Fe^{3+}$

Fe^{3+} : [Ar]



Not enough space for six H_2O molecule. Hence H_2O is not a strong ligand which may cause re-arrangement. Thus 4d orbital takes place in hybridization.

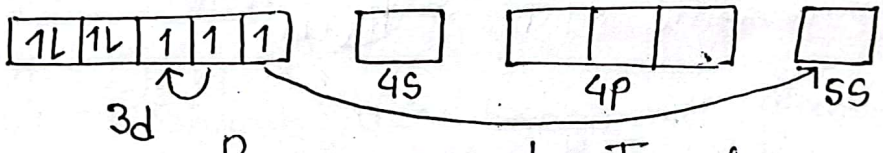
Fe^{3+} :



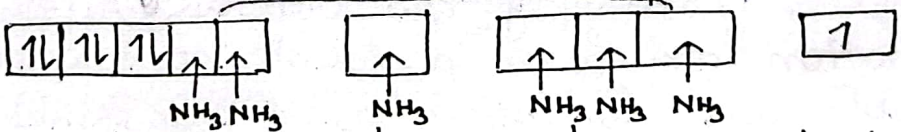
∴ Octahedral & strongly paramagnetic.

- $[Co(NH_3)_6]^{2+} \rightarrow Co^{2+}$

Co^{2+} : [Ar]



Re-arrangement + Transfer



* If the single e^- would move to 4d instead of 5s the hybridization would be dsp3d. which is not possible.

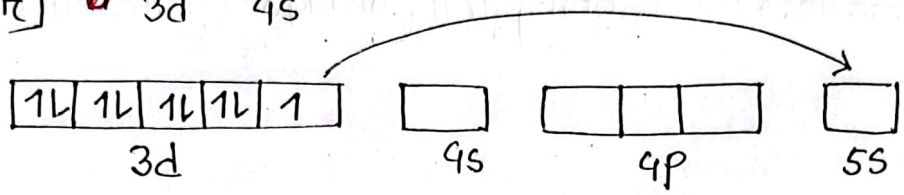
TOPIC NAME : _____ DAY : _____

TIME : DATE : /

- $[Cu(NH_3)_4]^{2+} \rightarrow Cu^{2+}$

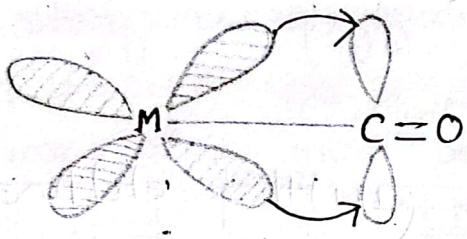
Cu (29) : [Ar] $3d^{10}\ 4s^1$

Cu^{2+} :



* If four NH_3 moves to 4s and 4p it would have hybridization sp^3 and tetrahedral shape. But experimentally square planare shape has been found. So, dsp^3.

- Back donation : (π bonding)



If only the ligand C=O. donates e^- to the metal, it would become unstable. To compansate metal also provide it's e^- to the empty P/d orbital of the ligand as back donation and form a double bond characterc. Due to double bond formation, metal gets stable.

- Condition for back donation-

(i) Metal should have filled d-orbital.

(ii) Ligand must contain empty P/d orbital.

(iii) Orbital symmetry of metal and ligand should be same.

[Farcia-50]

TOPIC NAME : DAY :
TIME : DATE : / /

□ What are the unique properties of transition metal elements ?

Ans:

Some unique properties of transition metal are-

(i) Have large charge/radius ratio.
(ii) Have high melting and boiling points
(iii) Show variable oxidation state
(iv) Form coloured ions and compounds
(v) Form compounds with profound catalytic activity.

□ Write down the nomenclature of the following complexes:

(i) $[Pt(NH_3)BrCl(CH_3NH_2)]$

Aminebromochloromethylamineplatinum (II).

(ii) $[Co(H_2NCH_2CH_2NH_2)_2Cl_2]^+$

Dichlorobis(ethylenediamine) cobalt (III)

(iii) $[Co(NH_3)_4(OH)_2]^{3+}$

Tetraaminedihydroxo cobalt (V)

(iv) $Mo[(C_6H_5)_2PCH_2CH_2P(C_6H_5)_2]_2(N_2)_2$

Bis(1,2-diphenylphosphinoethen) bis(dinitrogen) Molybdenum (0)

(v) $[Co(Co(NH_3)_4(OH)_2)_3]^{6+}$

Tris(tetraamine-μ-dihydroxo-cobalt) cobalt (6+).

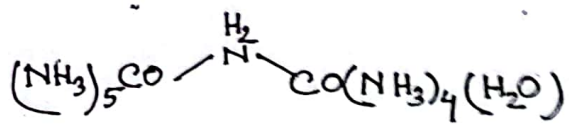
TOPIC NAME : _____ DAY : _____ TIME : DATE :

(vi) mer - $[Ru\{(C_2H_5)_3P\}_3H_3]$

mer-tris (triethylphosphine) trihydridoruthenium(III)

(vii) $(NH_3)_5CoNH_2Co(NH_3)_4(H_2O)]Cl_5$

Pentaammine cobalt(II)-μ-amino-tetraamme aquacobalt (III) chloride



(viii) $[Cr(H_2O)_4Cl_2]^+$

Tetraaqua dichloro chromium(III)

(ix) $[Co(NH_3)_5(SO_4)]^+$

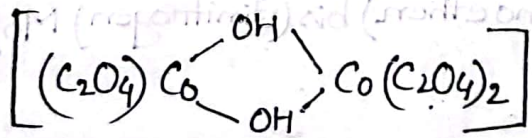
Pentaamminesulfato cobalt(III)

(x) $K_3[Al(C_2O_4)_3]$

Potassium tris(oxalato)aluminate(III)

(xi) $[Pt(NH_3)_4(NO_2)Cl]SO_4$

Tetraammine chloronitroplatinum ( ) Sulfate

(xii) K_4 

Potassium bis(μ_2-hydroxo bis(oxobato) Cobalt (III)

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Define denticity. When the eta (η) or kappa (κ) notations are used in nomanclature? Draw the structure of the following ligands and mention their denticity:

(i) Trien (ii) EDTA (iii) acac (iv) dppe.

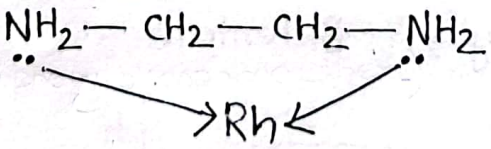
Ans:

Denticity is the number of donor groups or the number of pairs of shared electrons of the same ligand that binds with a metal atom or ion. Example: H_2O, NH_3 etc has only one donor atom, thus it's denticity is one.

When co-ordination of ligand is contiguous the eta (η) notation is used. Example,



When co-ordination of ligand is non-contiguous then we use kappa (κ) notation. Example:



TOPIC NAME : _____ DAY : _____ TIME : DATE : /

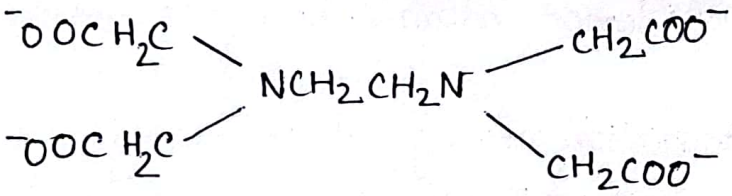
(i) Trien:

$NH_2CH_2CH_2NHCH_2CH_2NHCH_2CH_2NH_2$

As it contains, four nitrogen atom, it's denticity is four.

Denticity → Tetradentate.

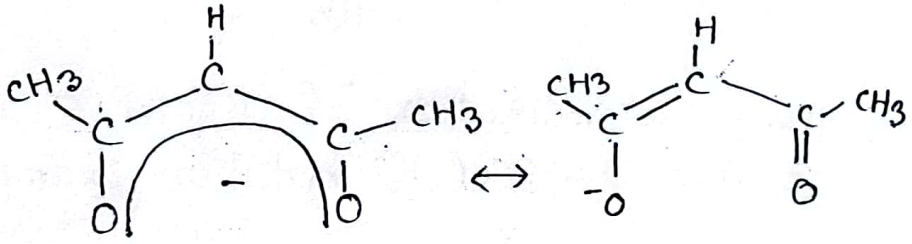
(ii) EDTA: Ethylene diammine tetraacetic acid



Denticity → Hexadentate

(iii) acac

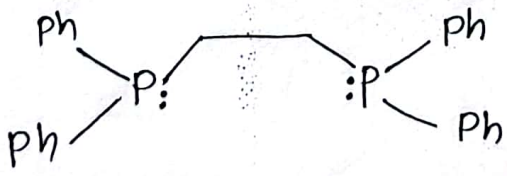
Acetyla-acetonato



Denticity → Bidentate

(iv) dppe:

(Diphenylphosphino) ethane



Denticity → Bidentate

TOPIC NAME : _____ DAY : _____

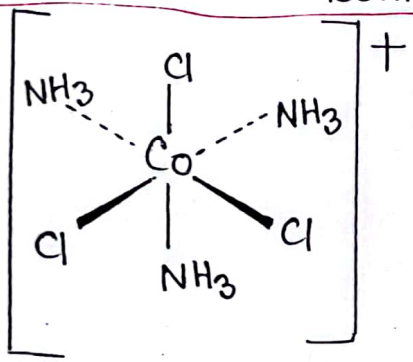
TIME : DATE : / /

□ Define isomerism. Draw structures of the fac and mer isomers of the complex $[Co(NH_3)_3Cl_3]$.

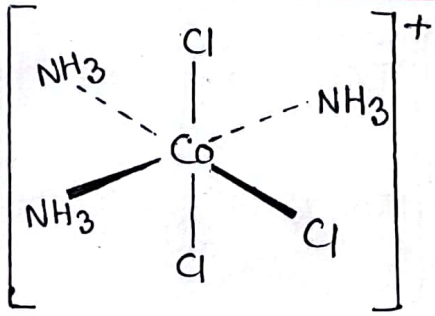
Ans:

Isomerism is a phenomenone used to describe complexes with the same molecular weight and molecular formula but different structural and/or spatial arrangement of donor atoms around the central metal in the co-ordination sphere.

The fac and mer isomers of the complex $[Co(NH_3)_3Cl_3]$ are



fac-triamminetrichlorocobalt (III)



mer-triamminetrichlorocobalt(III

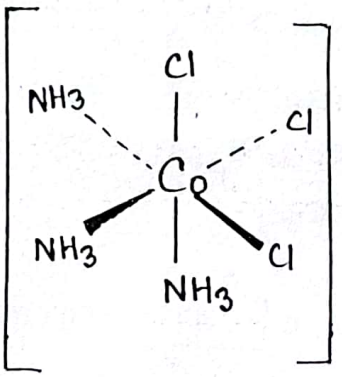
TOPIC NAME : _____ DAY : Farria-S

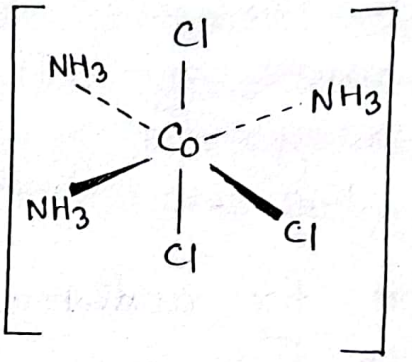
TIME : DATE : /

□ Draw the possible isomeric structures of $[Co(NH_3)$ and $[Co(dien)_2]^{3+}$

Ans:

The possible structures of $[Co(NH_3)_3Cl_3]$ are-

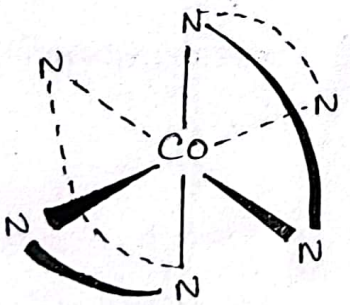




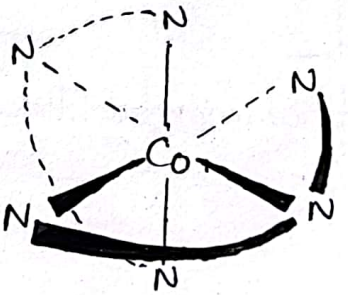
fac-triamminetrichlorocobalt(III)

mer-triamminetrichlorocobalt(III)

The possible isomeric structures of $[Co(dien)_2]^{3+}$ are-



Facial



Meridional

TOPIC NAME : _____ DAY : _____ TIME : DATE : / /

□ What is meant by stereochemically inert lone pair ? Give an example.

Ans:

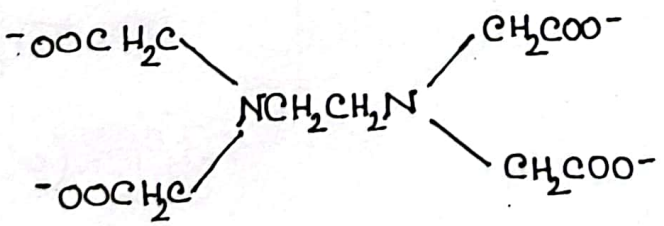
Stereochemically inert lone pairs are those that do not significantly influence the molecular geometry of a compound; i.e. they do not contribute to the spatial arrangement of atoms around a central atom.

Xe is a noble gas having eight valance electrons. In XeF_6, six of these electrons form bonds with fluorine atoms, leaving two as a lone pair. However, XeF_6 has regular octahedral geometry indicating that the lone pair doesn't influence the arrangments of F atom and termed as inert lone pair.

□ What do you understand by denticity of ligand ? Give example.

Ans:

Denticity refers to the number of donor atoms a ligand can use to bind to a central metal ion. EDTA is a hexadentate ligand meaning that it can bind to a metal ion using six donor groups/atoms : two nitrogen atoms and four oxygen atoms.



TOPIC NAME : _____ DAY : _____ TIME : DATE : /

TOPIC NAME • Isomerism

□ What is chelate effect ? Give example.

Ans:

The chelate effect is the increased stability of a metal complex formed by a ligand that can bind to a metal ion through multiple donor atoms to form a chelate ring. EDTA is a hexadentate ligand than can form a chelate ring complex with metal ions. It has a strong chelating effect due to its ability to form multiple five-membered chelate rings with the metal ion. by increasing the entropy of the system.

□ What are chelating ligand? Draw the structure of,
(i) Ca-EDTA (ii) $[Cr(oxalato)_3]^{3-}$

Ans:

Ligands with multiple donor sites that are able to form five to six-membered stable rings with metals are called chelating ligands.

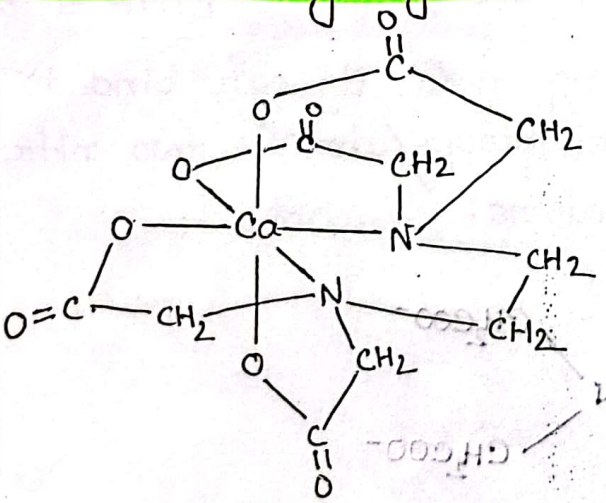


Fig: Ca-EDTA

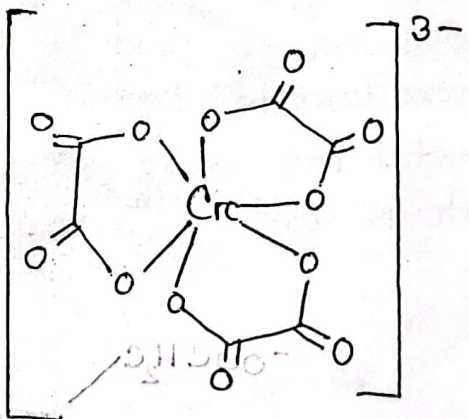


Fig: $[Cr(oxalato)_3]^{3-}$

TOPIC NAME : _____ DAY : _____ TIME : DATE : / /

• Isomerism :

Structural isomerism :

• Ionization isomerism :

The isomerism where pairs differ in that there is an exchange of two anionic groups within and outside the co-ordination sphere.

Ex : $[Co(NH_3)_5Br]SO_4$ & $[Co(NH_3)_5SO_4]Br$

$[Pt(en)_2Cl_2]Br_2$ & $[Pt(en)_2Br_2]Cl_2$

• Hydrate isomerism :

There are many isomers which differ in the position of water molecules in their formula.

Ex : $[Cr(H_2O)_6]Cl_3$ & $[Cr(H_2O)_5 \cdot Cl]Cl_2 \cdot H_2O$ &

$[Cr(H_2O)_4 \cdot Cl_2]Cl \cdot 2H_2O$

• Co-ordination isomerism :

Phenomenone where isomers contain pairs of ionic comple that exchange ligands with each other is called co-ordina isomerism.

Ex : $[Co(NH_3)_6][Cr(CN)_6]$ & $[Cr(NH_3)_6][Co(CN)_6]$

$[Co(en)_3][Cr(CN)_6]$ & $[Cr(en)_3][Co(CN)_6]$

CamScanner

TOPIC NAME : DAY : TIME : DATE :

• Linkage isomerism:

The type of isomerism observed in complexes containing ambidentate ligands which can co-ordinate through at least two different binding sites.

Ex: $[Co(NH_3)_5(NO_2)]^{2+}$ & $[Co(NH_3)_5(ONO)]^{2+}$

$[Co(CN)_5(SCN)]^{3-}$ & $[Co(CN)_5(NCS)]^{3-}$

• Polymerisation isomerism:

Polymerisation isomers contain the same simplest unit called monomer.

Ex: $[Pt(NH_3)_2Cl_2]$ unit on combination can give;

$[Pt(NH_3)_4][PtCl_4]$, $[Pt(NH_3)_3Cl]_2[PtCl_4]$ & $[Pt(NH_3)_4][Pt(NH_3)Cl_3]_2$

• Ligand isomerism:

Isomerism due to isomeric nature of ligands.

Ex: $[Pt(NH_2CH_2CH_2CH_2NH_2)_2]^{2+}$ & $[Pt(NH_2CH(CH_3)CH_2NH_2)_2]^{2+}$

Here the ligand propane 1,3-diamine and its isomer methylethylenediamine impose the isomerism.



TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

• Stereoisomerism :

Isomerism occurs in complexes due to difference in spatial arrangement of ligands round the central metal.

Q Draw the stereoisomers of $[Co(en)_2Cl_2]^+$ and linkage isomers of $[Co(NH_3)_5(NO_2)]^{2+}$

Ans:

Stereoisomers :

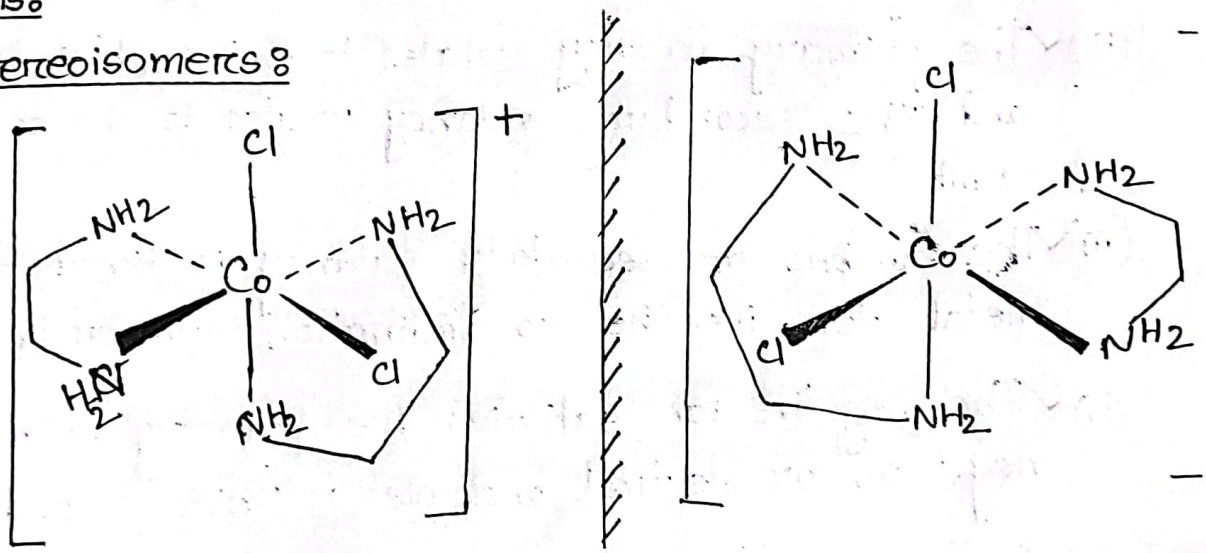


Fig: Cis-dichlorobis (ethylenediammine) Cobalt(III) Optical isomers (enantiomers).

~~Linkage isome~~

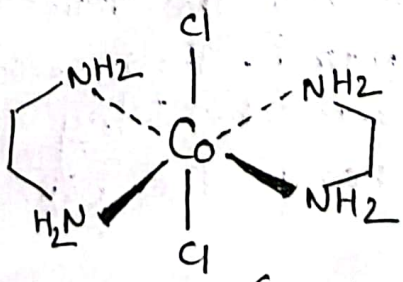


Fig: Trans-dichlorobis (ethylenediammine) cobalt(III) [Meso compound] Optically inactive

Linkage isomers :

$[Co(NH_3)_5(NO_2)]^{2+}$ and $[Co(NH_3)_5(ONO)]^{2+}$

TOPIC NAME : _____ DAY : Farida-50

TIME : DATE : / /

• Warnerr's Theory:

□ What are the postulates of Werrnerr's Theory ?

Ans:

Some important postulates of Werrnerr's Theory are-

(i) The central metal in the co-ordination compounds show two types of valency : Primary valancy & secondary valancy.

(ii) The primary valancy relates to the oxidation state and the secondary valancy relates to the coordinate numberr.

(iii) The numberr of secondary valancy is fixed forr any metal atom. i.e. the co-ordination numberr is fixed.

(iv) The negative ion satisfies the primary valancy and neg ion orr neutral molecule satisfies secondary valancy.

(v) The secondary valancy point towards a fixed position in space. This defines the definite geometry of the co-ordinate compound. Thus secondary valancy determines the sterreochemistry of the complex ion. On the otherr hand, the primary valancy is non-directional.

(vi) The fulfilment of secondary valancies appearr to be essential in complex formation.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ What are the limitations/drawbacks of Werner's Theo

Ans:

Some limitations of Werner's Theory:

(i) It could not explain the inability of all elements tha form coordination compounds.

(ii) The Werner's theory could not explain the directio properties of bonds in various co-ordination compou

(iii) It does not explain the color, the magnetic and optic properties shown by co-ordination compounds.

□ Define primary and secondary valancies with example

Ans:

• Primary valancy: Primary valancy is the number of neg ions that are directly bonded to a central metal ion tha satisfy its charge. This type of bonding is typically ionisa & so bonds can be dissociated in solution. They are non-directional in nature. Primary valancy is equal to the oxidation state of metal ion. In complex $[Co(NH_3)_6]Cl_3$ primary valancy is 3.

• Secondary valancy: Secondary valancy is the number ligands directly bonded to metal ion through co-ordi bond. These ligands can be neutral or ions. It repres the spatial arrangement of ligands around the metal it is typically fixed for a given metal. In complex $[Co(NH_3)_6]Cl_3$ the secondary valancy is six.

CamScanner

TOPIC NAME : _____ DAY : _____

TIME : DATE : /

□ Differentiate between primary and secondary valan

Ans8

Primary valancy	Secondary valancy
1. Related to metals oxidation state.	1. Related to co-ordination numberc of metal.
2. Usually satisfied by ionic bonds	2. Involves co-ordinate co-valent bonds.
3. It's variable	3. Fixed in numberc.
4. Can be dissociated in soln.	4. Usually remains undissociate
5. Not involved in determining the geometry of complex	5. Determines the geometry the complex.
6. Non-directional	6. Directional.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

• Ligand :

Q1 Define ligand. How ligands are classified in co-ordinati chemistry ? Give an example for each type.

Ans:

A ligand is an ion or molecule that binds to a central metal atom to form a co-ordinate complex. The bindin with ligand involves formal donation of one or more electron pairs from the ligand.

Ligands can be classified based on the number of don sites they posses :

(i) Monodentate ligand : Has one donor site. eg. H_2O, NH_3

(ii) Bidentate ligand : Has two donor site. Ex: Ethylenediamin ($NH_2CH_2CH_2NH_2$), oxalate $(C_2O_4)^{2-}$ etc.

(iii) Tridentate ligand : Has three donor site. Ex: Diethylenetria $(NH_2CH_2CH_2NHCH_2CH_2NH_2)$.

(iv) Quadridentate : Has four donor site. Ex: Triethylenete $(NH_2CH_2CH_2NHCH_2CH_2NHCH_2CH_2NH_2)$.

(v) Polydentate : Has many donor site. Ex: EDTA

(vi) Amidentate : (Another category) They have two hetero different donor and can be used to co-ordinate with metal depen on the reaction conditions.

Ex: SCN^- can co-ordinate either with S or N.

NO_2^- " " " " N or O.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ What are tripodal ligands ? Give one specific exam of this type of ligands.

Ans:

Tripodal ligands are a class of polydentate ligands that have three pronged, tripod-like structure. They typically contain three donor atoms (such as O, N/S) that are arranged in a way that allows them to co-ordinate to a metal ion simultaneously.

Nitrilotriacetate $[N(CH_2CO_2H)_3]$ is a type of tripodal ligand which has three acetate groups attached to a central nitrogen atom.

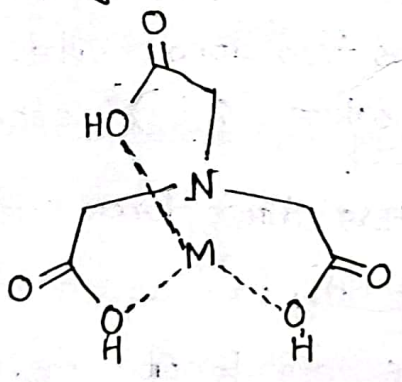


Fig: Metal nitrilotriacetate.

Some examples are:

Nitrilotriacetate – $N(CH_2COOH)_3$

Atriamine – $N(CH_2CH_2NH_2)_3$

A triphosphine – $RC(CH_2PPh_2)_3$

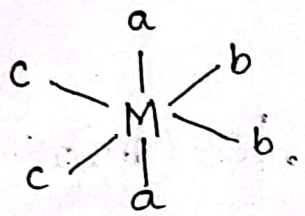
TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

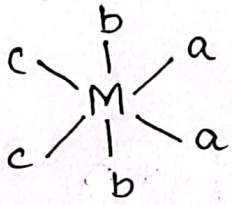
□ Find the number and identity of all the isomers $[Ma_2b_2c_2]$

Ans:

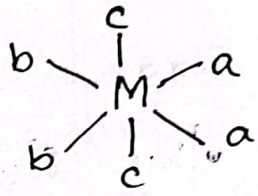
Five geometrical isomers by formula $Ma_2b_2c_2$.



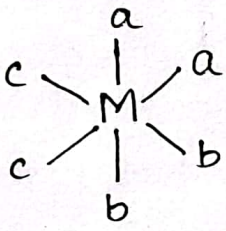
a-a trans



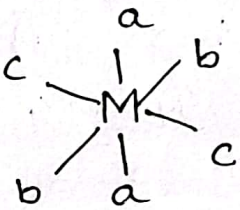
b-b trans



c-c trans

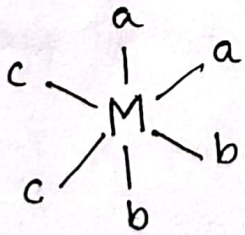


all cis



all trans

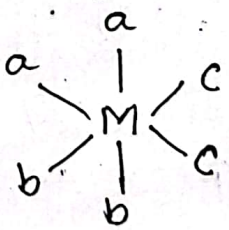
Two optically active isomers.



cis-d-isomer



Mirror



cis-l-isomer

Optically active

TOPIC NAME : _____ DAY : _____

TIME : DATE : /

□ Calculate EAN for the central metal atom of the following complexes :

$[Ru(CO)_3(PPh_3)_2]$, $[V(CO)_6]$. $[Co(NH_3)_4(NO_2)Cl]Cl$,

$K_3[Fe(CN)_5(NO)].2H_2O$ $[Co(NH_3)_5(NO_2)]$

Ans :

• $[Ru(CO)_3(PPh_3)_2] \rightarrow Ru^0$

$Ru \rightarrow e^- \rightarrow 44$

From three $CO \rightarrow e^- \rightarrow 6$

" two $PPh_3 \rightarrow e^- \rightarrow 4$

EAN = 44+6+4 = 54

• $V[CO]_6 \rightarrow V^0$

$V \rightarrow e^- \rightarrow 23$

EAN = 23 + 2X6 = 35

• $[Co(NH_3)_4(NO_2)Cl]Cl \rightarrow Co^{3+}$

$Co^{3+} \rightarrow e^- \rightarrow (27-3) = 24$

EAN = 24 + 2X4 + 2X1 = 36

• $K_3[Fe(CN)_5(NO)].2H_2O \rightarrow Fe^{2+}$

$Fe^{2+} \rightarrow (26-2) = 24$

EAN = 24 + 5X2 + 1X2 = 36

• $[Co(NH_3)_5(NO_2)] \rightarrow Co$

$\Rightarrow Co^+ \rightarrow e^- \rightarrow (27-1) = 2$

EAN = 26 + 12 = 38.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ What is the notion of VBT proposed by Linus Pauling in explaining structure of co-ordination complexes ?

Ans :

Pauling's VBT :

(i) The central metal cation or atom makes avaiable nu of vacant s, p, d orbitals equal to its co-ordination m to form co-ordinate covalent bonds with ligands.

(ii) These vacant atomic orbitals of metal are hybridized form a new set of equivalent bonding orbitals calle hybrid orbitals. These orbitals have the same geometry the same energy and definite directional properties.

(iii) The bonding in metal complexes arises when a fil ligand orbital containing a lone pair of electron.

(iv) Pauling classified the ligands into two categories strong ligand and weak ligand.

(v) Strong ligand have the tendency to pair up the delocalized electron of a metal cation or atom to the necessary orbitals for hybridization. On the other hand, weak ligands do not show such kind of tend

(vi) The d orbitals used in hybridization may be either inner (n-1)d orbitals or outer d-orbitals.

CamScanner

TOPIC NAME : _____ DAY: _____

TIME : DATE : /

(vii) If unpair electron is present, then the complex is paramagnetic otherwise, they are called diamagne

Limitations :

(i) It could not explain the number of ligand why the pairing of e⁻ occurs in presence of ligands. color and electronic spectra of complexes.

(ii) It does not provide any thermodynamic and kinetic stability of complexes.

□ What are the differences between hapticity and Dentis

Hapticity	Denticity
1. Hapticity is the coordination of a ligand to a metal center via uninterrupted and contiguous series of atoms.	1. Denticity is the number of donor groups of the same ligand that binds with a metal center.
2. Gives the number of contiguous atoms involved in the coordination process.	2. Gives the number of donor groups of the ligand attach with the metal center.
3. η - notation	3. K notation.
4. Ferrocene has two ligands bound to Fe center and each ligand binds through 5 contiguous atoms.	4. EDTA coordinates with metal via 6 non-contiguous atoms..

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

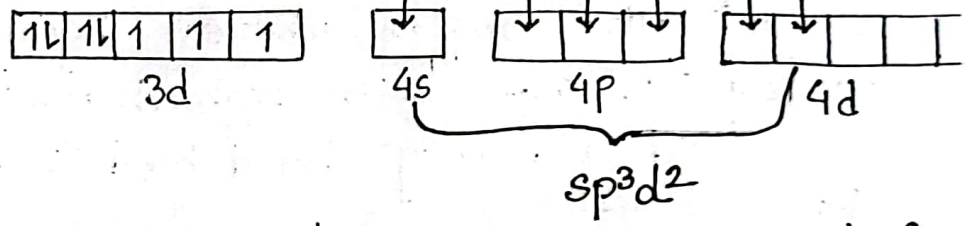
□ Describe the σ bonding in the following complexes us VBT. mention their hybridization and molecular struc

(i) High spin $[Co(NH_3)_6]^{3+}$ (ii) $[CoCl_4]^{2-}$ (iii) $[Ni(CN)_4$

Ans:

(i) $[Co(NH_3)_6]^{3+} \rightarrow Co^{3+}$

Co^{3+}



sp^3d^2

As the complex compound is high spin compound. So re-arrangement of orbitals doesn't occur. Hence, the hybridization is sp^3d^2.

Geometry → Octahedral.

[(ii) & (iii) have been discussed earlier.]

□ Why the transition metal have higher hydration energy?

Ans:

The transition metal ions are small in size and have high charge. Due to this, the size/charge ratio is low and charge/size ratio is high. Thus hydration energy is usually high for transition metal.

TOPIC NAME : _____ DAY : _____

TIME : DATE : /

* □ Lability of the complex decreases with the increa of charge on metal ion – explain.

Ans:

Lability refers to how easily metal ligand bonds are broken. A complex compound in which metal ion function as an electrophile or lewis acid in forming co-ordination complex, whereas ligand function as a nucleophile. Metal with high charges act as a strong lewis acid and they should bind ligands more tightly As a result -

(i) Metal ion with higher charges and small size form stable complexes.

(ii) A small cation with high charge attracts the ligands more closely leading to stable complex.

□ Explain -" Geometrical isomerism is not possible for tetrahedral complex. "

Ans:

Geometrical isomer requires possibility of a different structure where the distance between any two particular groups can be changed as compared to first structure.

CamScanner

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Tetrahedral complexes doesn't show geometrical isome

(i) All the four ligands are equidistant from one another.

(ii) The relative position of ligand attached to the ce metal atom are same with respect to each other

(iii) All bond angles are 109.5°. Thus there is no s of one atom being next to and another beir across trans.

□ Some Name of Co-ordinate complexes:

- $[Cu(NH_3)_4]SO_4$ – Tetraammine copper (II) sulfate
- $Na_3[Fe(CN)_6].2H_2O$ – Sodium hexacyanoferrate(III) 2-w
- $[Pt(acac)(NH_3)Cl]$ – Acetylacetonatoammine chloroplatinum (II)
- $[Os(CO)_3(PPh_3)_2]$ – Tricarbonyl bis (triphenylphosphine) Osmiur

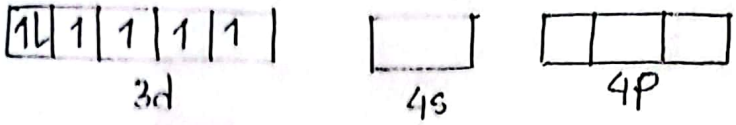
TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

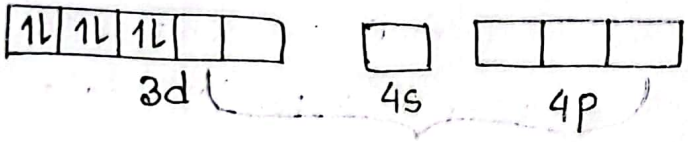
□ $[Fe(CN)_6]^{4-}$ is diamagnetic but $[FeF_6]^{3-}$ is paramag Explain.

Ans 8 $[Fe(CN)_6]^{4-}$

The oxidation state of the Fe is Fe^{2+}.

Fe^{2+} : 

with the prescence of CN^-, re-arrangement of Fe^{2+} occures.

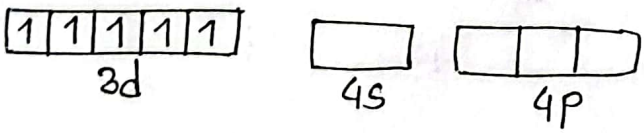
Fe^{2+}: 

d^2sp^3

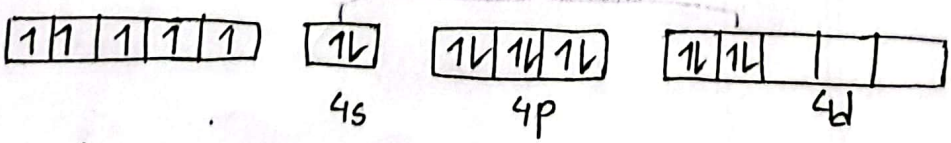
$[Fe(CN)_6]^{4-}$: 

There is no unpaired electron. Thus it is diamagnet

$[FeF_6]^{3-} \rightarrow Fe^{3+}$

Fe^{3+} : 

F^- being a weak ligand can not re-arrange the electron Thus 4d orbital also takes place in hybridization to accomodate six fluorides. sp^3d^2

$[FeF_6]^{3-}$: 

As, it has five unpaired electron it is paramagnetic.

[Faria-50]